Motaxle

Preliminary ERPNext and Jinchen MES Implementation Estimate

Status: Preliminary Estimate
Date: 29 September 2026

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1. Overview

This document provides a preliminary estimate for implementing **ERPNext for Motaxle**, including integration with **Jinchen MES**.

The estimate is based on the information currently available. Since the detailed forging/fabrication workflow, costing methodology, quality process and Jinchen integration requirements are still pending, the timeline has been kept conservative.

Estimated Overall Effort

Approximately 242–324 working days

This is a preliminary estimate and should be refined once the detailed requirements and Jinchen integration specifications are available.

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2. Current Understanding

Based on the information available:

- Motaxle is a manufacturing/engineering company working with forging, fabrication and motion-related components.
- ERPNext is being considered as the ERP platform for the greenfield implementation.
- SAP Business One and SAP S/4HANA are also being considered.
- Jinchen MES will be involved in shop-floor manufacturing execution.
- ERPNext and Jinchen MES will need to exchange manufacturing-related information.
- The detailed Motaxle manufacturing workflow has not yet been received.

Because of this, the current estimate should be treated as a **preliminary planning estimate**.

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3. Preliminary Scope

3.1 Discovery & Business Analysis

- Detailed forging/fabrication process mapping
- Plant and warehouse structure
- Item and UOM structure

- BOM requirements
- Routing and operations
- Production planning and MRP
- Procurement and subcontracting
- Quality process
- Batch/heat/lot/serial traceability
- Costing methodology
- Jinchen MES integration requirements

Estimated effort: 23–30 working days

3.2 ERPNext Core Implementation

- Company setup
- Users, roles and permissions
- Sales and order management
- Procurement and supplier management
- Inventory and warehouse management
- BOM
- Routing
- Workstations
- Production planning
- Work Orders
- Job Cards
- Quality management
- Subcontracting/job work, if required
- Accounts and financial transactions
- Costing and taxes

Estimated effort: 53–68 working days

3.3 Manufacturing / Forging Customization

Potential customization areas:

- Forging/fabrication-specific workflows

- Production operation-specific fields and statuses
- Costing customizations
- Heat/lot traceability
- Tooling, die and fixture tracking
- Tool-life tracking, if required
- Rework workflows
- Rejection and scrap handling
- NCR-related workflows
- Production reports and dashboards

Estimated effort: 45–60 working days

The final effort will depend heavily on the actual Motaxle manufacturing process.

3.4 Inventory, Procurement & Costing

Potential areas include:

- Raw material management
- Purchase processes
- Warehouse movements
- Material consumption
- Material returns
- Subcontracting/job work
- Scrap and wastage
- Standard/actual costing
- Machine and labour costs
- Overhead allocation
- Cost variance reporting

Estimated effort: 30–38 working days

3.5 Quality, Traceability & Reports

- Incoming quality inspection
- In-process quality checks
- Final inspection

- Rejection and rework
- Quality parameters
- Batch/heat/lot traceability
- Serial tracking, if required
- Production reports
- Quality reports
- Management dashboards

Estimated effort: 23–30 working days

3.6 Jinchen MES Integration

The integration may include:

- Integration architecture
- API and field mapping
- Production Order / Work Order exchange
- BOM and routing information
- Material issue and movement
- Material returns
- Production confirmation
- Finished Goods receipt
- Rejection and rework
- Quality results
- Traceability information
- Error handling
- Retry mechanism
- Integration logs
- Reconciliation
- Integration monitoring

Estimated effort: 53–75 working days

The available Jinchen reference shows similar integration patterns such as work orders, BOMs, material movements, finished goods receipt, rework and traceability. However, that reference is for another manufacturing business and should not be treated as Motaxle's confirmed flow.

3.7 Testing, UAT & Deployment

- Functional testing
- Integration testing
- End-to-end manufacturing testing
- UAT support
- Data migration support
- Production deployment
- Initial stabilization

Estimated effort: 15–23 working days

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4. Overall Estimated Timeline

Workstream	Estimated Time
Discovery & Business Analysis	23–30 days
ERPNext Core Implementation	53–68 days
Manufacturing / Forging Customization	45–60 days
Inventory, Procurement & Costing	30–38 days
Quality, Traceability & Reports	23–30 days
Jinchen MES Integration	53–75 days
Testing, UAT & Deployment	15–23 days
Total Estimated Effort	**242–324 days**

These are **working-day estimates for the individual workstreams**. The actual project calendar can be shorter if multiple workstreams are handled in parallel.

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5. Costing Considerations

The costing requirement needs further clarification. Depending on Motaxle's process, the system may need to account for:

- Raw material
- Purchased components
- Forging

- Cutting
- Heating
- Fabrication
- Machining
- Heat treatment
- Surface treatment
- Machine hours
- Setup time
- Labour
- Consumables
- Tooling and dies
- Fixtures
- Tool-life costs
- Scrap
- Rejection
- Rework
- Subcontracting
- Factory overhead

The final costing approach should be confirmed with Motaxle before the implementation estimate is finalized.

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6. Key Dependencies

The following information is still required:

- Detailed Motaxle forging/fabrication workflow
- Production routing and operations
- BOM structure
- Quality checkpoints
- Traceability requirements
- Costing methodology
- Tooling/die requirements
- Jinchen MES API documentation

- Exact data that needs to flow between ERPNext and Jinchen
- Data migration requirements
- Final reporting requirements

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7. Main Estimation Risks

Area	Possible Impact
Manufacturing workflow not finalized	Additional customization may be required
Costing methodology not finalized	Additional costing logic may be required
Jinchen API details unavailable	Integration effort may change
Traceability requirements unclear	Data model and workflows may change
Machine/shop-floor integration unclear	Additional integration work may be required
Data migration not assessed	Migration effort may increase

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8. Conclusion

ERPNext can be considered for the Motaxle project, particularly since this is a greenfield implementation. However, the current information is not sufficient to finalize the scope or timeline.

For the current planning stage, the estimated effort is **approximately 242–324 working days**.